Aggregate Fluctuations with Incomplete Information and Heterogeneous Risk*

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Abstract

Evidence of individual income *heteroskedasticity*, i.e., heterogeneous income risk, is pervasive in micro data. We conjecture an incomplete information framework with Bayesian learning as a plausible microfoundation for individual income heteroskedasticity. We study the determinants of aggregate fluctuations and the monetary and fiscal policy mulipliers in a heteroskedastic New Keynesian model. We obtain three main results. First, marginal propensities to consume (MPCs) are heterogeneous, as in the data. They result from the heterogeneity in the *sensitivity* of individual risk and precautionary savings to realized earnings. Second, the response of aggregate output to demand shocks hinges crucially on how individual *estimated* risk co-varies with the degree of individual income cyclicality across the income distribution. Third, the general equilibrium effects of monetary and fiscal policy can be suitably summarized by a set of observable cross-sectional sufficient statistics. Conditional on realistic signs of those statistics, income heteroskedasticity amplifies the response of output to demand shocks.

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1 Introduction

A recent literature in macroeconomics has developed models that incorporate heterogeneity and uninsurable idiosyncratic risk into the standard New Keynesian (NK) apparatus with a representative agent (aka HANK models). There is, however, a large debate on *which* dimension of heterogeneity is essential to deepen our understanding of aggregate fluctuations as well as the channels of transmission of monetary and fiscal policy.

Traditionally, the literature on heterogeneous agent models has been centered on the assumption that individual households face a *homoskedastic* income process: that is, the variance of the income process is homogeneous in the cross-section and constant over time. This is at odds with the evidence from micro data that points to significant cross-sectional *heteroskedasticity*: in particular, individual income risk is tied to past realizations of individual income (Guvenen et al., 2021; Arellano et al., 2021). However, the relation between the volatility of the income process and the income distribution has been largely ignored in the recent HANK literature.

We provide a microfoundation for the link between individual income volatility and lagged earnings, based on *incomplete information*. We assume that households cannot observe the volatility of their income process and use the available information to make inference on the unknown variance parameter. In particular, we impose heterogeneity in the information set across agents, since each household observes both the aggregate variables and the realization of her own income process, but *not* those of other households. Each individual's current earnings realization is the private signal used to update the estimate of future volatility, via a Bayesian learning procedure. Therefore, heterogeneity in the *signals* translates into heterogeneous *estimates* of the variance of the income process. As a result, heteroskedasticity in the individual income process can be derived as an implication of informational frictions and heterogeneity in the households' information set.

Recent papers in the HANK literature have shown that the *cyclicality* of income risk, i.e., how individual risk behaves during expansions and contractions, can be a source of either amplification (in the case of counter-cyclical risk) or dampening (in the case of pro-cyclical risk) of aggregate fluctuations.[1] For instance, agents react to negative aggregate demand shocks by decreasing their consumption relatively more when risk is counter-cyclical, since aggregate contractions are also associated with higher individual risk and precautionary savings. While this notion of risk-cyclicality captures the idea that idiosyncratic income risk may be *on average* higher during recessions,[2] it disregards the evidence that risk varies *across the earnings distribution*.

Within our setting, realized changes in individual earnings act as signals through which agents revise their *perceived* risk, and thus their optimal consumption and savings decisions. Suppose, consistent with our empirical evidence, that the estimated variance of the individual income process is an inverse (and non-linear) function of realized earnings. In this case, a negative realization of

[1] For instance, Acharya and Dogra (2020) and Bilbiie (2018) for income risk, Ravn and Sterk (2021) for unemployment risk.

[2] See, e.g., Storesletten et al. (2004).

income also increases individual estimated risk, thereby strengthening at the margin the agent's motive for precautionary savings. As a result, the *sensitivity* of precautionary savings to income realizations will be heterogeneous across individuals.

We employ our incomplete-information heteroskedastic NK framework to study the transmission of aggregate demand (monetary and fiscal) shocks. We reach three main results.

First, incomplete information acts as a *microfoundation* for heteroskedasticity. In particular, heterogeneity in private signals, i.e, in information sets across agents, translates into heterogeneous perceived individual risk, which agents learn in a Bayesian fashion.

Second, and unlike the homoskedastic particular case, our setup with heteroskedasticity features a non-degenerate *distribution* of MPCs out of transitory income. This is due to realized earnings affecting estimated income volatility at the margin, and therefore also precautionary savings and consumption choices, as long as the marginal change in volatility varies over the earnings distribution. Specifically, our model is able to generate MPCs which are decreasing over the income distribution, in line with the empirical evidence.[3] To gain an intuition for this result, suppose that perceived income volatility is decreasing and convex in realized earnings. Then any additional unit of income determines a decline in precautionary savings that is, at the margin, stronger for low-income households, who in turn react by spending more of the income shock. As a result, they exhibit relatively higher MPCs.

Third, the presence of heteroskedasticity has general equilibrium implications that are absent both in a representative-agent (RA) model and in a corresponding version of our framework with homoskedasticity. The interaction between estimated income risk and the income distribution determines indirect effects that shape the responses of key aggregate variables to demand shocks. More specifically, the general equilibrium indirect effects of heteroskedasticity can be conveniently summarized in terms of three *sufficient statistics*: (1) the expected marginal change in estimated income volatility out of realized individual earnings, capturing an *individual risk* effect; (2) the average sensitivity of individual income to aggregate income, capturing an *income cyclicality* effect; (3) the covariance between (1) and (2), i.e., between the marginal change in income volatility and the sensitivity of individual income to aggregate income, capturing a *risk-cyclicality covariance* effect .

Those sufficient statistics allow to reduce the dimensionality of the model when describing the dynamic response of output to demand shocks. In particular, statistics (1) and (2) jointly capture how, *on average*, perceived risk varies as a function of individual income in response to variations in aggregate output. Statistic (3), on the other hand, captures the extent to which perceived individual risk *varies differently across agents* depending on how individual income is affected by aggregate income fluctuations.

[3] Jappelli and Pistaferri (2014) estimate (using survey data from the Bank of Italy Survey of Household Income and Wealth) that annual MPCs are 11 percent higher in the lowest quintile of the income distribution relative to the highest quintile. Johnson et al. (2006) also find (with CEX data) that MPCs out of the 2001 U.S. tax rebates are higher for low-income households.

Depending on the signs of the aforementioned sufficient statistics, heteroskedasticity could either amplify or dampen the response of output to demand shocks. In particular, negative (positive) signs for statistics (1) and (3) amplify (dampen) the effect of demand shocks on output.[4] Conversely, both in the benchmark complete-market RA model and in the homoskedastic version of our framework, statistics (1) and (3) are constant and equal to zero at all times.

Notice that statistics (1) and (2) jointly describe the size of the effect on output of *average* changes in risk. For instance, and following a negative demand shock, a stronger average increase in income volatility generates a larger increase in precautionary savings, and therefore a larger contraction in consumption. Statistic (3), on the other hand, captures the *interaction* between income risk and income redistribution over the business cycle. Suppose that a negative demand shock has a first-round *direct* contractionary effect on output. To start with, lower aggregate income affects each agent depending on the heterogeneous sensitivity of individual income to aggregate income. In our heteroskedastic setup, however, any change in individual earnings affects also the volatility of the income process, and therefore saving and consumption decisions. Suppose the covariance term (3) is, realistically, negative. Then, those agents whose income falls relatively more out of a contraction in aggregate output are also the ones who experience a marginally stronger increase in idiosyncratic risk. At the margin, the same agents exhibit a relatively stronger precautionary saving motive and therefore will decrease their consumption relatively more. In other words, those agents whose income is more heavily affected by the contraction in aggregate output are also the ones who have a stronger incentive to decrease their consumption. This indirect *propagation through-risk* channel generates an amplification of the primitive shock on aggregate output.

In the last section of the paper we describe a full general equilibrium model with distortions on the supply side (nominal price rigidities and market power). We compute analytically the fiscal policy and monetary policy multipliers. We show that, to the extent that individual perceived risk increases more for those agents whose income is more cyclical, heteroskedasticity acts as an amplifier of monetary and fiscal shocks.

1.1 Relation to the literature

Our work relates to several strands of the literature. First of all, the paper relates to a new stream of the HANK literature which is moving away from the assumptions of rational expectations and perfect information. Laibson et al. (2021) study a HANK model where agents have present bias and naive beliefs. Auclert et al. (2020) estimate a HANK model with sticky household expectations. Pfäuti and Seyrich (2022) analyze how cognitive discounting and household heterogeneity affect the transmission of monetary policy. While we mantain the hypothesis of rational expectations, we introduce imperfect information and heterogeneity of signals in a context with Bayesian learning.

[4] In general, statistic (2) – the average sensitivity of individual to aggregate income – is positive, since average individual income is generally increasing in output.

Within the perfect information HANK literature, Kaplan et al. (2018) disentangle *direct* and *indirect* effects of monetary policy on output and show that, in a HANK model, policy transmission works mainly through the indirect channel. We highlight a new general equilibrium channel of transmission of monetary and fiscal policy, stemming from the interaction between income redistribution and changes in income volatility. Recent papers (such as Auclert, 2019; Auclert et al., 2018; Debortoli and Galí, 2022; Wolf, 2021) have highlighted the role of sufficient statistics as a way to reduce the dimensionality of HA models and to summarize the equilibrium effects of fiscal and monetary policy. We contribute further by showing that the heteroskedasticity channel of output responses to demand shocks can be suitably captured by a few observable sufficient statistics.

There is a large literature on the theory of income risk and market incompleteness. In particular, within a CARA utility framework, Caballero (1990) shows how income risk produces a drift in the consumption process leading to excess consumption growth. Wang (2002) enriches the CARA model by Caballero (1990) to study the role of conditionally heteroskedastic income, postulating a linear relation between conditional income volatility and lagged earnings. Compared to those papers, our model, while making use of CARA utility, imposes only a differentiability requirement to the conditional variance (as a function of lagged earnings), without restricting the dependence to be linear. Wang (2003) and Acharya and Dogra (2020) combine the consumption model of Caballero (1990) with a general equilibrium setting. We contribute further to this section of the literature by showing the general equilibrium effects of heteroskedasticity.

The HANK setup of Bayer et al. (2019) assumes a heteroskedastic process for labor productivity where the evolution of the variance over time depends on exogenous shocks. Unlike them, we focus on the cross-sectional dimension and we take heteroskedasticity as endogenously determined by the income distribution. Bilbiie (2018) builds a two-agent NK model augmented with idiosyncratic risk, modeled through a Markov transition structure between hand-to-mouth (HtM) and non-HtM states where the transition probabilities co-move with aggregate output. Since non-HtM agents are identical across them, they all face the same individual risk and aggregate fluctuations are linked to the cyclical property of the transition probabilities. In addition, Acharya and Dogra (2020) build a CARA utility model with a Gaussian income process where the volatility is identical across households, but depends over time on aggregate output, capturing the cyclicality of individual income risk. Unlike these papers, we obtain that individual income is endogenously heteroskedastic in the cross-section and we show that cross-sectional heterogeneity in idiosyncratic risk interacts with the earnings distribution, thereby affecting aggregate fluctuations in response to demand shocks.

Debortoli and Galí (2022) study the effects of heterogeneous changes in individual consumption risk over the business cycle. Their setup has two main features. First, a precautionary motive due to incomplete insurance. Second, heterogeneity in MPCs. In their model, this last feature is due to heterogeneous assets positions, hence to a heterogeneous distance from a natural borrowing limit. Differently from them, in our model heterogeneity in MPCs is rooted in heteroskedasticity.

Finally, we contribute to the literature on *tractable* heterogeneous agent models (for instance,

Acharya et al., 2020; Acharya and Dogra, 2020; Bilbiie, 2008, 2018, 2020; Bilbiie et al., 2013; Challe, 2020; Challe and Ragot, 2016; Ravn and Sterk, 2021) building a general equilibrium HA framework where we integrate heteroskedasticity and idiosyncratic risk with heterogeneity in MPCs and a non-degenerate wealth distribution in an analytically tractable and transparent way.

The rest of the paper is structured as follows. Section 2 describes the empirical evidence on cross-sectional heteroskedasticity. Section 3 presents the model. Section 4 concludes.

2 Evidence on heteroskedasticity: effective vs. perceived risk

A large empirical evidence supports the hypothesis of cross-sectional heteroskedasticity in the individual income process.[5] Moreover, recent contributions in the literature on non-parametric estimation of the income process show the existence of a relation between the volatility of the income process and the percentiles of the lagged income distribution.[6] Guvenen et al. (2021) find evidence of an asymmetric U-shaped relation between the standard deviation of the two-year log-earnings growth and the percentiles of the lagged earnings distribution.[7] In particular, the standard deviation is decreasing and convex up to the 95th percentile and sharply increasing for higher percentiles. Arellano et al. (2021) also study the moments of the income process, with Spanish administrative data, using an approach akin to the one in Guvenen et al. (2021). They find that the standard deviation of two-year log-earnings growth is decreasing over the percentiles of lagged income.

Consistent with this literature, we provide evidence of heterogeneity in the standard deviation of log-earnings across the earnings distribution. We use data from the waves of the Panel Survey of Income Dynamics (PSID) during the period 1969-2019 to estimate the standard deviation of the income process over lagged income bins.[8] We focus on labor earnings of family heads aged between 25 and 65, dropping low-earnings observations (annual earnings less than $1,000 in 2012 dollars). Our observations are at a biannual frequency.

To estimate income volatilities, for each year, we first split observations in deciles of lagged earnings.[9] Second, we filter log-earnings through a constant and a linear time trend, both specific to each lagged income decile.[10] Then, we compute (for each year in our sample) the standard deviation of filtered log-earnings conditional on the deciles of the lagged earnings distribution. Finally, we average the estimated standard deviations (for each lagged earnings bin) over time.

Figure 1 shows the estimated standard deviation of the income process, for each decile of the

[5] See Meghir and Pistaferri (2004).

[6] See Arellano (2014), Arellano et al. (2017), Arellano and Bonhomme (2017), and Guvenen et al. (2014).

[7] Guvenen et al. (2021) employ a large panel dataset drawing a representative 10% sample of the U.S. population from the Social Security Administration (SSA), and measure the presence of non-linearities and non-normalities in individual earnings dynamics over the life cycle. They document the existence of a significant empirical relation between higher moments of the income process (i.e., standard deviation, skewness, kurtosis) and age/percentiles of the income distribution.

[8] See Appendix A.1 for a more detailed description of the data.

[9] We define lagged earnings as earnings two years earlier.

[10] We also condition on age and gender.

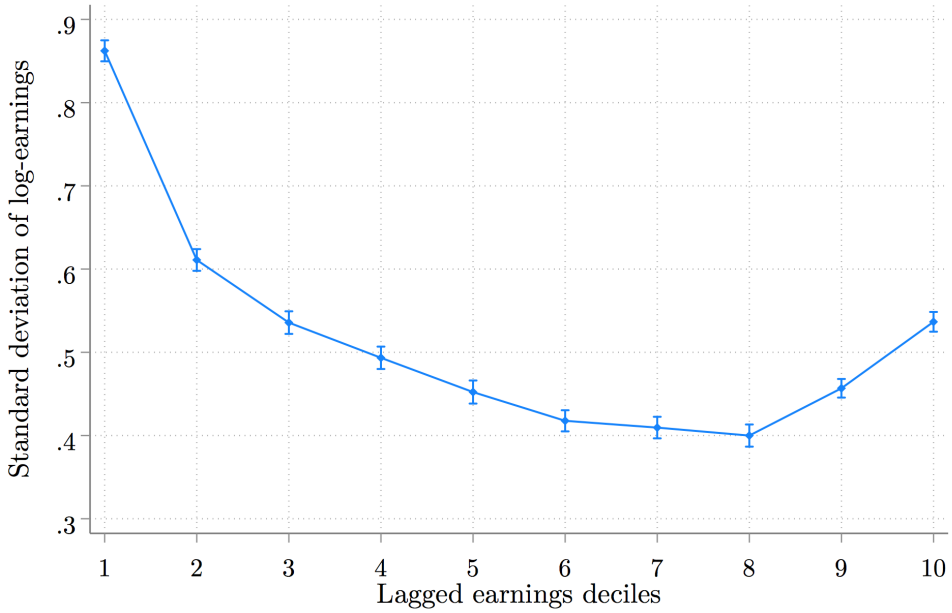

FIGURE 1: ESTIMATES OF THE STANDARD DEVIATION OF LOG-EARNINGS

Notes. The figure shows the standard deviation of filtered log-earnings, for each decile of the distribution of (two-years) lagged earnings. Filtered log-earnings are the residuals of a regression of log-earnings on a constant and a linear time trend, specific to each lagged earnings decile (controlling also for age and gender indicators). We report confidence intervals at the 90% level.

(lagged) earnings distribution. The observed pattern is similar to the one found in Guvenen et al. (2021) and based on administrative data.[11] The volatility of the income process is heterogeneous and depends on the position of a given individual over the earnings distribution. In particular, the standard deviation of log-earnings is decreasing up to the 8th decile of the earnings distribution and increasing over the last two deciles. Moreover, the decline in the standard deviation over the earnings distribution is large in magnitude: the standard deviation is halved as an individual moves from the 1st to the 8th decile.

The evidence above describes an inverse relationship between *effective* risk and individual income. As we argue throughout the paper, under imperfect information, income risk may not be observable by the agent, so that income risk is only *perceived*.

2.1 Perceived heteroskedasticity

While the evidence above points to heterogeneity in *effective* income risk, recent survey data provide evidence of heterogeneity in the *perception* of idiosyncratic income risk over the earnings

[11] See Figure C.1 (left panel) in the Online Appendix in Guvenen et al. (2021).

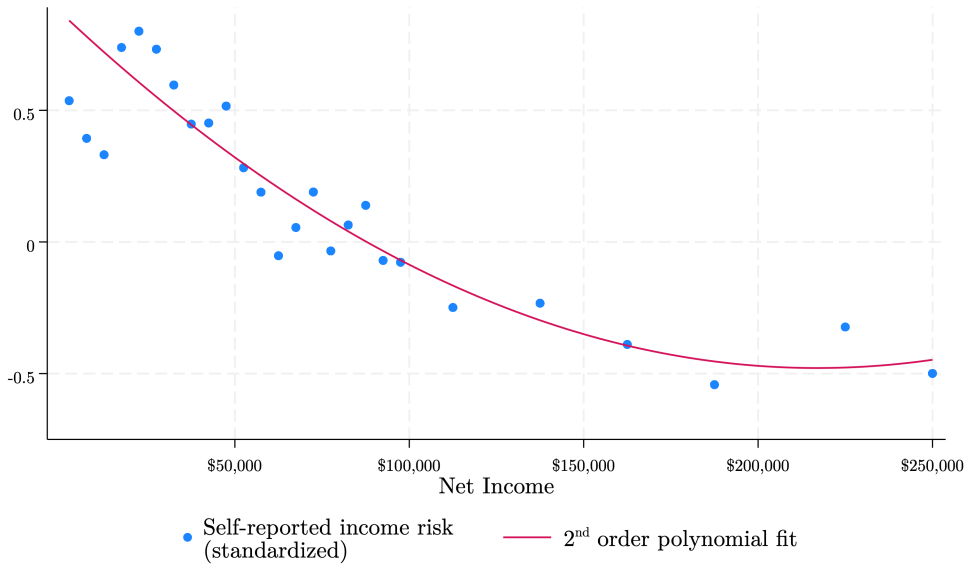

FIGURE 2: HETEROGENEITY IN PERCEIVED INCOME RISK OVER THE EARNINGS DISTRIBUTION

Notes. The graph plots the self-reported income risk across gross earnings bins (blue dots). We fit the data with a 2nd order polynomial. Figure from Colarieti, Mei, and Stantcheva (2023).

distribution.[12] Colarieti, Mei, and Stantcheva (2023) run a large-scale survey of U.S. households[13] to estimate macroeconomic moments and to study how people think about the macroeconomic environment. A question in their survey asks respondents to evaluate how risky they perceive their future income.[14]

Figure 2 shows the existence of a stark relation between perceived income risk and earnings:[15] perceived risk is decreasing over large part of the income distribution and flat for the last two income bins.[16] The magnitude of the change in perceived risk over the earnings distribution is also meaningful: households with annual net income of less than $50,000 perceive their risk to be 1 standard deviation higher than those with income greater than $150,000.

To summarize, both a measure of *actual* risk and a measure of *perceived* risk relate inversely to income across the income distribution. Below we propose an explanation for this relationship based on imperfect information. Perceived risk is heterogeneous across agents because each agent uses realized income as a private signal to estimate the unobserved variance of the income process. Het-

[12]For instance, Ben-David et al. (2018) show that income is negatively correlated with uncertainty expectations. They build an index taking into account self-reported uncertainty - asking respondents to attribute subjective probabilities to different scenarios - of one-year personal income growth, national inflation and national house-price growth, therefore combining macro and micro uncertainty.

[13]Respondents are individuals in the labor force, aged between 25 and 65 years old.

[14]The formulation of the question is: "How certain or uncertain are you about your total household income over the next 12 months?" [Extremely certain; Very certain; Somewhat certain; Neither certain nor uncertain; Somewhat uncertain; Very uncertain; Extremely uncertain].

[15]Earnings are measured as self-reported household total annual gross income.

[16]The relation is robust to residualizing the measure of self-reported income risk with demographic and occupation controls (age, race, education, household size, occupation, employment status).

erogeneity in information sets therefore translates into heterogeneity in signals and, consequently, heterogeneity in estimated income risk.

3 A heteroskedastic HANK model with incomplete information

In this section we lay out a HANK model where agents feature *incomplete information*. We assume that agents cannot observe the variance of the true income process and, in a Bayesian sense, must make inference on this moment using the realizations of their income as private signals. As a result, changes in realized earnings induce households to revise the estimates of their income volatility, modifying their *perception* of individual risk. In turn, this leads agents to revise their optimal consumption and saving decisions.

In this setting, the fundamental income process is assumed to be homoskedastic (in the cross-section). However, it is the *combination* of incomplete information *and* heterogeneity in signals (for income realizations differ across agents) that determines heterogeneity in the estimated volatility.

3.1 General state-space formulation

Below we lay out a general state-space formulation for the evolution of the volatility of the income process.

3.1.1 Model and information structure

The economy is populated by a continuum of households in the interval $[0,1]$, indexed by $i \in [0,1]$. We assume a *state-space* model where the volatility σ_t^2 of individual income (homoskedastic in the cross-section) is the state variable, behaving according to a Markov process. Individual income is the *observation* variable, assumed to be i.i.d. across agents and independent over time, conditional on the current variance state. Formally:

Assumption 1. *For every period t the fundamental individual income process for agent i is described by the following state-space formulation:*[17]

$$\sigma_t^2 \sim G_t(d\sigma_t^2 \mid \sigma_{t-1}^2), \tag{1a}$$

$$y_{i,t} \sim F_t(dy_{i,t} \mid \sigma_t^2), \tag{1b}$$

where G_t is the probability transition kernel[18] *for the variance (i.e., the state variable) from σ_{t-1}^2 to $d\sigma_t^2$, and F_t is the probability measure for individual income, conditional on the realization of time t variance σ_t^2.*

[17] We work here with probability measures, hence the notation $d\sigma_t^2, dy_{i,t}$.

[18] More formally, by definition of transition kernel, $G_t : (\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+)) \to (\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$ is a function such that $G_t(d\sigma_t^2 \mid \sigma_{t-1}^2)$ is a probability measure on $(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$ (where $\mathcal{B}(\mathbb{R}_+)$ is the Borel sigma-algebra on the set of positive real numbers) for every σ_{t-1}^2 and the map $\sigma_{t-1}^2 \mapsto G_t(A \mid \sigma_{t-1}^2)$ is measurable for every set $A \in \mathcal{B}(\mathbb{R}_+)$.

Notice that both G_t and F_t are common across agents, which in turn implies that the individual income process is identical across agents. In addition, the above state-space formulation requires the observation variable to be conditionally independent of past observations.

We postulate that each agent knows the state-space model in (1) and can observe the history of the earnings realizations and aggregate variables. However, the agent does not observe neither the realizations of the variance σ_t^2 (which needs to be estimated) nor other agents' earnings. In other words, we assume that the individual income process is a *private signal* that each household exploits to forecast her future income volatility, a key moment in the determination of her optimal consumption and saving decision. While income volatility evolves over time according to a Markov process, agents cannot observe it and use the history of current and past individual income realizations to obtain forecasts for the future variance of the income process.

3.1.2 Learning

We assume that rational agents *learn* over time about the variance of the income process in a Bayesian way, using the history of their earnings to derive state-predictive posterior distributions of future income volatility. Let $y_{i,0:t} := \left(y_{i,k}\right)_{k=0}^{t}$ denote the *history* of individual income realizations at time t (i.e., the vector of past and current realizations from 0 to t). Then, the output of agents' learning in every period t can be described by two conditional distributions:

- The *filtering distribution*: $\mathbb{P}_t(d\sigma_t^2 \mid y_{i,0:t})$, i.e., the distribution of the current (time t) income volatility conditional on the earnings history at time t.
- The *state-predictive distribution*: $\mathbb{P}_t(d\sigma_{t+1}^2 \mid y_{i,0:t})$, i.e., the distribution of future (time $t+1$) income volatility conditional on the earnings history at time t.

To obtain the above mentioned distributions, the Bayesian learning procedure can be described by the following algorithm. Let $i \in [0,1]$ be a generic agent in the economy.

Initial stage, $t = 0$. Each agent i has a common prior $\mathbb{P}_0(d\sigma_0^2)$ over σ_0^2. After observing her first-period income $y_{i,0}$, agent i computes the filtering distribution through Bayesian updating:

$$\mathbb{P}_0(d\sigma_0^2 \mid y_{i,0}) \propto F_0(dy_{i,0} \mid \sigma_0^2) \cdot \mathbb{P}_0(d\sigma_0^2), \tag{2}$$

obtaining the state-predictive distribution for σ_1^2 as:

$$\mathbb{P}_0(d\sigma_1^2 \mid y_{i,0}) = \int_{\sigma_0^2 \in \mathbb{R}_+} G_1(d\sigma_1^2 \mid \sigma_0^2) \cdot \mathbb{P}_0(d\sigma_0^2 \mid y_{i,0}). \tag{3}$$

Recursive stages, $t \geq 1$. For the following periods, the learning procedure can be described recursively. At time $t \geq 1$, the prior is given by the state-predictive distribution computed by the agent in $t-1$, i.e., $\mathbb{P}_{t-1}(d\sigma_t^2 \mid y_{i,0:t-1})$.

When income is drawn at time t from the conditional distribution $F_t(dy_{i,t} \mid \sigma_t^2)$, agent i updates her prior, obtains the filtering distribution and, in turn, the new state-predictive distribution. In particular, the filtering distribution is given (up to a normalization constant) by:[19]

$$\mathbb{P}_t(d\sigma_t^2 \mid y_{i,0:t}) \propto F_t(dy_{i,t} \mid \sigma_t^2) \cdot \mathbb{P}_{t-1}(d\sigma_t^2 \mid y_{i,0:t-1}), \tag{4}$$

while the new state-predictive distribution is given by:

$$\mathbb{P}_t(d\sigma_{t+1}^2 \mid y_{i,0:t}) = \int_{\sigma_t^2 \in \mathbb{R}_+} G_{t+1}(d\sigma_{t+1}^2 \mid \sigma_t^2) \cdot \mathbb{P}_t(d\sigma_t^2 \mid y_{i,0:t}). \tag{5}$$

The above equation illustrates our key point. Namely, that the state-predictive distribution depends on the individual earnings history. This implies that agents who have experienced heterogeneous income realizations over time also derive heterogeneous distributions of future income volatility. This establishes a relationship between agent i's *estimated distribution* of the income variance and agent i's position in the earnings distribution. The existence of this link has, in turn, implications for each agent's optimal consumption and savings decisions: households with different earnings histories have different perceptions of their individual income risk and, therefore, make heterogeneous consumption choices.

3.1.3 Consumer problem under imperfect information

The agent can purchase, in every period, a risk-free real bond $a_{i,t+1}$ (paying one unit of good in $t+1$) at the price $1/(1+r_t)$ and consume a perishable consumption good $c_{i,t}$, using its total (net) income $y_{i,t}$ and its beginning-of-period wealth $a_{i,t}$. The recursive formulation of the consumer's problem requires to keep track of the filtering and the state-predictive distributions:

$$\begin{aligned} V_t(a_{i,t}, y_{i,t}) := \max_{(c_{i,t}, a_{i,t+1})} & \left\{ u(c_{i,t}) + \beta \cdot \mathbb{E}_{i,t}\left[V_{t+1}(a_{i,t+1}, y_{i,t+1})\right] \right\} \\ \text{s.t.} \quad & c_{i,t} + \frac{a_{i,t+1}}{1+r_t} = y_{i,t} + a_{i,t}, \quad a_{i,0} \text{ given.} \end{aligned}$$

Notice that the expected value on the right-hand side of the Bellman equation is computed with respect to the distribution of future earnings conditional on the individual earnings history:

$$\mathbb{E}_{i,t}\left[V_{t+1}(a_{i,t+1}, y_{i,t+1})\right] = \int_{y_{i,t+1} \in \mathbb{R}} V_{t+1}(a_{i,t+1}, y_{i,t+1}) \cdot \mathbb{P}_t(dy_{i,t+1} \mid y_{i,0:t}),$$

[19] This expression is derived from Bayes' Theorem and from the assumption of conditional independence of income, i.e., $\mathbb{P}_t(dy_{i,t} \mid \sigma_t^2, y_{i,0:t-1}) = F_t(dy_{i,t} \mid \sigma_t^2)$:

$$\begin{aligned} \mathbb{P}_t(d\sigma_t^2 \mid y_{i,0:t}) &= \frac{\mathbb{P}_t(dy_{i,t} \mid \sigma_t^2, y_{i,0:t-1})}{\mathbb{P}_t(dy_{i,t} \mid y_{i,0:t-1})} \cdot \mathbb{P}_{t-1}(d\sigma_t^2 \mid y_{i,0:t-1}) \propto \\ &\propto F_t(dy_{i,t} \mid \sigma_t^2) \cdot \mathbb{P}_{t-1}(d\sigma_t^2 \mid y_{i,0:t-1}). \end{aligned}$$

where:

$$\mathbb{P}_t(dy_{i,t+1} \mid y_{i,0:t}) = \int_{\sigma^2_{t+1} \in \mathbb{R}_+} F_{t+1}(dy_{i,t+1} \mid \sigma^2_{t+1}) \cdot \mathbb{P}_t(d\sigma^2_{t+1} \mid y_{i,0:t}),$$

and $\mathbb{P}_t(d\sigma^2_{t+1} \mid y_{i,0:t})$ is given by equation (5). Importantly, the consumer problem is recursive: agent i does not need to know her earnings history in every period t, but can simply carry over her state-predictive distribution as a state variable. In particular, taking $\hat{\mathbb{P}}_t(d\sigma^2_t) := \mathbb{P}_{t-1}(d\sigma^2_t \mid y_{i,0:t-1})$ as a state variable in t, equations (4) and (5) describe the law of motion to obtain $\hat{\mathbb{P}}_{t+1}(d\sigma^2_{t+1}) = \mathbb{P}_t(d\sigma^2_{t+1} \mid y_{i,0:t})$ through the new income observation $y_{i,t}$.

3.2 An analytically tractable setup

We assume that the economy runs for two periods: $\tau = t, t+1$. This allows a closed-form solution to the consumption problem outlined above, so that the implications of heterogeneity in private income signals on agents' optimal consumption decisions can be transparently laid out.

Compared to the general state-space model described above, we introduce a set of simplifying assumptions.

Assumption 2. *The true variance of the income process is constant over time:* $\sigma^2_{t+1} = \sigma^2_t = \sigma^2$.

Assumption 3. *The fundamental income process (that is i.i.d. across agents and independent across time) follows a Gaussian distribution (conditional on the variance* σ^2*):*

$$y_{i,t} \mid \sigma^2 \sim \mathcal{N}(y_t, \sigma^2),$$

where y_t *is the period* t *mean.*

As in the state-space formulation described above, agents know both the process followed by their individual income and its mean y_t, yet not its second moment σ^2. Therefore, they use their private income signal to estimate the variance of income.

Each agent starts with a (common) prior distribution[20] $\sigma^2 \sim p(\sigma^2)$ at time t and, after observing her current income realization $y_{i,t}$, she updates her prior through Bayesian learning to obtain the posterior distribution: $\sigma^2 \mid y_{i,t} \sim p(\sigma^2 \mid y_{i,t})$. More compactly:

$$y_{i,t} \mid \sigma^2 \sim \mathcal{N}(y_t, \sigma^2) \quad i.i.d., \tag{6a}$$

$$\sigma^2 \sim p(\sigma^2), \tag{6b}$$

where (6a) indicates that, conditional on a given volatility, income follows a Gaussian distribution, and (6b) represents the prior held by the agent on the income variance.

The posterior distribution is proportional to the prior times the likelihood function:

$$p(\sigma^2 \mid y_{i,t}) \propto f_t(y_{i,t} \mid \sigma^2) \cdot p(\sigma^2), \tag{7}$$

[20] We assume that the probability distributions admit densities. We use lower-case letters to denote densities, while capital letters to denote probability measures.

where f_t is the Gaussian density of $y_{i,t}$ (as well as the likelihood function for any given $y_{i,t}$). Notice that equation (7) is a simplified version of equation (4).

Each agent i estimates the income volatility as the expected value of the posterior distribution. More formally, we define the estimator of the variance as:

$$\hat{\sigma}_i^2 := \mathbb{E}_{i,t}(\sigma^2 \mid y_{i,t}) = \int_{\mathbb{R}_+} \sigma^2 \cdot p(\sigma^2 \mid y_{i,t}) \cdot d\sigma^2, \tag{8}$$

where $\mathbb{E}_{i,t}(\cdot)$ denotes the expectation operator conditional on agent i's information set at time t.

Notice that the estimates of income volatility are *heterogeneous* across agents with different earnings draws. In other words, heteroskedasticty is an *outcome* of the learning procedure.

Finally, we assume that each agent i uses the forecast $\hat{\sigma}_i^2$ *as if* it were the *correct* variance of the income process. This allows us to preserve the normality of the income process and to find closed-form solutions to the consumption problem.[21] It is therefore possible to define the relevant income process for consumption and saving decisions: $y_{i,t+1} \mid \mathcal{I}_{i,t} \sim \mathcal{N}(y_{t+1}, \hat{\sigma}_i^2)$ where $\mathcal{I}_{i,t}$ is agent i's information set at time t. In other words, after concluding their learning procedure, agents take their estimated future volatility as the true one and, conditional on their current income realization $y_{i,t}$, they know that their future income follows a Gaussian density with variance given by the estimated one.

3.3 Consumer problem and aggregate demand

Let $u(\cdot)$ be a CARA - exponential utility function: $u(c) = -\frac{1}{\gamma}\exp\{-\gamma c\}$ with γ being the coefficient of absolute risk aversion. The solution to the two-period optimal consumption problem is characterized by the following Euler equation:

$$\exp\{-\gamma c_{i,t}\} = \beta(1 + r_t) \cdot \mathbb{E}_t\big[\exp\{-\gamma c_{i,t+1}\}\big]. \tag{9}$$

the sequence of budget constraints, the initial condition on assets, and the transversality condition on bonds:

$$a_{i,t+1} = 0$$

The tractability of the CARA utility framework allows to derive[22] optimal current consumption explicitly as a function of the real interest rate, permanent income, and the estimate of income volatility:

$$c_{i,t} = \underbrace{-\frac{1}{(2+r_t)\gamma}\log[\beta(1+r_t)]}_{\text{intertemporal substitution}} + \underbrace{\frac{1+r_t}{2+r_t}\left(a_{i,t} + y_{i,t} + \frac{1}{1+r_t}y_{t+1}\right)}_{\text{permanent-income}} \underbrace{-\frac{1}{2+r_t}\frac{\gamma}{2}\hat{\sigma}_i^2}_{\text{estimated risk}}. \tag{10}$$

[21] Otherwise we would have to work with a non-Gaussian distribution for unconditional income, and we would not be able to obtain closed-form solutions.

[22] See Appendix B.1 for derivations.

The first term on the right-hand side of equation (10) captures a typical intertemporal substitution motive stemming from real interest rate movements, while the second term captures the implications of the permanent-income hypothesis (i.e., consumption in every period is a fraction of lifetime wealth). Finally, the last term represents a shifter that depresses current consumption and captures the effects of *estimated* risk on precautionary savings. In particular, if the agent estimates higher values for income volatility (higher $\hat{\sigma}_i^2$), then she optimally increases her precautionary savings and reduces current consumption.

The MPC out of wealth reads:

$$MPC_{i,t}^{\text{wealth}} = \mu_t = \frac{\partial c_{i,t}}{\partial a_{i,t}} = \frac{1+r_t}{2+r_t}, \tag{11}$$

MPCs out of wealth are therefore homogeneous across agents. The MPC out of transitory income shocks is:[23]

$$MPC_{i,t} = \frac{\partial c_{i,t}}{\partial y_{i,t}} = \mu_t - \frac{1}{2+r_t}\frac{\gamma}{2}\frac{\partial \hat{\sigma}_i^2}{\partial y_{i,t}}. \tag{12}$$

It is clear from the previous expressions that if the marginal change in the estimated income volatility out of current earnings (i.e., $\partial\hat{\sigma}_i^2/\partial y_{i,t}$) is non-null and non-constant across the earnings distribution, then the MPCs out of *income* (i) differ across agents, (ii) are time-varying, and (iii) differ from the MPC out *wealth*.

The mechanism that produces heterogeneity in MPCs is as follows. Under incomplete information, changes in income are equivalent to changes in the *signal* received by each agent, who will therefore revise her own estimate of income volatility. Suppose, for instance, that in response to a marginal increase in her current earnings agent i revises downward her estimate of the income variance relatively more compared to agent j. Then agent i will perceive her future income as being relatively less *risky*, and will reduce her precautionary savings (increase her consumption) relatively more at the margin: therefore agent i will feature a relatively higher MPC out of transitory income.

3.3.1 Aggregation and general equilibrium

After aggregation of the consumption functions in (10) and imposing market clearing $y_t = c_t + g_t$, where g_t denotes government consumption, the aggregate demand function can be obtained as:[24]

$$y_t = y_{t+1} - \frac{1}{\gamma}\log[\beta(1+r_t)] - \frac{\gamma}{2}\int_{[0,1]}\hat{\sigma}_i^2 di + g_t - g_{t+1}. \tag{13}$$

As we can see, as a result of a precautionary saving motive, average *perceived* income volatility depresses aggregate output.

[23] Assume that $\hat{\sigma}_i^2 : \mathbb{R} \to \mathbb{R}_+$ is differentiable with respect to income realizations $y_{i,t}$.

[24] See Appendix B.2 for derivations.

3.3.2 Sufficient statistics

We are now ready to establish one of our key results. We show that the equilibrium response of output to alternative demand shocks can be fully described through a small set of observable sufficient statistics, capturing the heterogeneous consumption choices of different agents.

Proposition 1. *The equilibrium response of aggregate output to government spending and interest rate shocks is captured by the following cross-sectional sufficient statistics:*

(i) Expected marginal change in the estimate of income volatility out of earnings:

$$\underbrace{\mathbb{E}\left(\frac{\partial \hat{\sigma}_i^2}{\partial y_{i,t}}\right)}_{\textit{individual risk effect}}.$$

(ii) Expected earnings sensitivity to aggregate output:

$$\underbrace{\mathbb{E}\left(\frac{\partial y_{i,t}}{\partial y_t}\right)}_{\textit{income cyclicality effect}}.$$

(iii) Covariance between the marginal change in the estimate of income volatility and the earnings sensitivity to aggregate output:

$$\underbrace{Cov\left(\frac{\partial \hat{\sigma}_i^2}{\partial y_{i,t}}, \frac{\partial y_{i,t}}{\partial y_t}\right)}_{\textit{risk-cyclicality covariance effect}}.$$

Let Θ_t be the output multiplier out of a fully transitory government spending shock (assuming that the real interest rate is unchanged) :

$$\Theta_t := \frac{\partial y_t}{\partial g_t} = \left\{1 + \frac{\gamma}{2}\left[\mathbb{E}\left(\frac{\partial \hat{\sigma}_i^2}{\partial y_{i,t}}\right) \cdot \mathbb{E}\left(\frac{\partial y_{i,t}}{\partial y_t}\right) + Cov\left(\frac{\partial \hat{\sigma}_i^2}{\partial y_{i,t}}, \frac{\partial y_{i,t}}{\partial y_t}\right)\right]\right\}^{-1}. \tag{14}$$

Moreover, the multplier out of a transitory monetary shock (e.g., an increase in the real interest rate) reads:

$$\frac{\partial y_t}{\partial r_t} = -\Theta_t \cdot \frac{1}{\gamma(1 + r_t)}. \tag{15}$$

Proof. See Appendix B.3.

Consider statistics (i)-(iii). All combined they capture how individual *perceived risk* varies over the business cycle and how this effect is heterogeneous across the income distribution.

Statistic (i) captures the expected marginal effect of realized income on individual perceived risk (*individual risk effect*). Statistic (ii) captures the degree of cyclicality of individual income with respect to aggregate income (*income cyclicality effect*). Finally, statistic (iii) describes how the effect on individual risk co-varies with the degree of income cyclicality (*risk-cyclicality covariance effect*). Hence, the size of both the government spending and real interest rate multipliers depends on the interaction between the cross-sectional variation in the precautionary saving motive and how the distribution of earnings is shaped by aggregate output.

Consider the (product of the) first two sufficient statistics in (14), i.e., (i) the individual risk effect and (ii) the income cyclicality effect. That product is likely to be negative. Suppose that a negative demand shock decreases output on impact and that, in turn, individual earnings decline, i.e., the income cyclicality effect is positive.[25] If, *on average*, individual perceived risk is decreasing in individual income, as documented in Figure 2, then the product of (i) and (ii) is negative. Hence, following the negative demand shock, agents face *more risk* on average and therefore feature a larger precautionary saving motive. Therefore, they decrease consumption relatively more than if average risk were unaffected.

However, the latter precautionary savings response is *not homogeneous* across agents. This can be gauged from statistic (iii), which captures the risk-cyclicality covariance effect. The latter statistic describes the interaction between how individual perceived risk varies at the margin and how sensitive earnings are to aggregate output over the income distribution.

Intuitively, consider a negative demand shock. As aggregate income decreases, individual earnings may react in the cross-section in a heterogeneous fashion. Then, changes in individual earnings affect future income volatility, and in turn precautionary savings and individual MPCs. In turn, the heterogeneous consumption response at the margin feeds back into aggregate demand. By way of example, suppose that idiosyncratic perceived risk is decreasing in individual income, i.e., $\partial\sigma^2_{i,t+1}/\partial y_{i,t} < 0$. If those agents whose income is more sensitive to aggregate income are also those for whom perceived risk increases (at the margin) relatively more (following a decrease in their individual earnings), then the covariance term is negative, leading to amplification. In other words, amplification of a negative demand shock requires that those agents whose income is more sensitive to aggregate income are also those with a stronger increase in the precautionary saving motive and higher MPCs.

Overall, then, *amplification* in the response of output to demand shocks requires a negative value for the covariance statistic and/or a negative value for the product between the average change in the estimate of risk and the average income sensitivity.

Consider, once again, a negative demand shock that has a first-round *direct* negative effect on aggregate output. Then, the reduction in output affects individual earnings in a heterogeneous fashion, depending on income sensitivities $(\partial y_{i,t}/\partial y_t)_{i\in[0,1]}$. As agents observe modifications in their signals, they revise their estimates of income volatility and change their consumption at the margin.

[25] That the income cyclicality effect is positive is consistent with empirical evidence in Patterson, 2021

For instance, if those agents featuring larger income sensitivities (i.e., those who lose relatively more from the contraction in output) are also those who revise their risk estimates upward marginally more, the decline in output will be amplified. In general, the interaction between revisions in estimates and income sensitivities shape an *indirect* effect of demand shocks on aggregate output.

3.4 A tractable case: Inverse-Gamma prior

A key advantage of the incomplete information setting as a foundation of heteroskedasticity is that it allows a flexible parameterization of the prior distribution of income volatility. In turn, this allows us to easily obtain closed-form solutions for the posterior estimator $\hat{\sigma}_i^2$.

We assume that each agent $i \in [0,1]$ has a common Inverse-Gamma prior, with common hyper-parameters $\sigma_0^2 > 0$, $\nu_0 > 0$:

$$\sigma^2 \sim \text{Inv-Gamma}\Big(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\Big). \tag{16}$$

After observing their income realization $y_{i,t}$, agents update the prior distribution following Bayes' Theorem and obtain the posterior distribution, which is itself an Inverse-Gamma with newly derived parameters:[26]

$$\sigma^2 \mid y_{i,t} \sim \text{Inv-Gamma}\Big(\frac{\nu_0 + 1}{2}, \frac{\nu_0 \sigma_0^2 + (y_{i,t} - y_t)^2}{2}\Big). \tag{17}$$

Then, each household estimates her own income volatility using the conditional expected value:

$$\hat{\sigma}_i^2 = \frac{\nu_0 \sigma_0^2 + (y_{i,t} - y_t)^2}{\nu_0 - 1}. \tag{18}$$

The intuition for (18) is standard in Bayesian statistics. The posterior estimator can be expressed as a weighted average of the prior mean[27] and the squared deviation of the income realization from its average, where the weights are given by the prior sample size and the actual sample size:[28]

$$\hat{\sigma}_i^2 = \Bigg[\frac{\nu_0 - 2}{\nu_0 - 1} \cdot \underbrace{\frac{\nu_0 \sigma_0^2}{\nu_0 - 2}}_{\text{prior mean}} + \frac{1}{\nu_0 - 1} \cdot \underbrace{(y_{i,t} - y_t)^2}_{\text{squared deviation}}\Bigg]. \tag{19}$$

In our model the actual sample size is equal to 1, since households observe a sample with only one (i.e., their own) income realization $y_{i,t}$. Instead, the prior sample size is $n_0 = \nu_0 - 2$. Therefore we can use the hyperparameter ν_0 to determine the prior sample size, while the prior mean is a function of both σ_0^2 and ν_0.

Let $\mathcal{Y}_{i,t} := y_{i,t} - y_t$ denote the deviation of realized individual income from its mean. Consider

[26] See Appendix B.4.

[27] We define the prior mean as the expected value of the variance with respect to the prior distribution:

$$\text{prior mean} := \mathbb{E}(\sigma^2).$$

See Appendix B.4 for details on the moments of the Inverse-Gamma distribution.

[28] We assume that $\nu_0 > 2$ for the existence of the first moment of the prior distribution.

the expression of the posterior estimator in (18). What actually matters for the estimate of the income variance is not income itself, but its deviation from its known mean (that coincides with aggregate output). Hence, we can rewrite the expression for the government spending multiplier Θ_t as:

$$\Theta_t := \frac{\partial y_t}{\partial g_t} = \left\{1 + \frac{\gamma}{2}\left[\mathbb{E}\left(\frac{\partial \hat{\sigma}_i^2}{\partial \mathcal{Y}_{i,t}}\right) \cdot \mathbb{E}\left(\frac{\partial \mathcal{Y}_{i,t}}{\partial y_t}\right) + Cov\left(\frac{\partial \hat{\sigma}_i^2}{\partial \mathcal{Y}_{i,t}}, \frac{\partial \mathcal{Y}_{i,t}}{\partial y_t}\right)\right]\right\}^{-1}. \tag{20}$$

Then, given the estimator $\hat{\sigma}_i^2$ in (18), we can derive explicit formulas for the sufficient statistics of Proposition 1:

$$\mathbb{E}\left(\frac{\partial \hat{\sigma}_i^2}{\partial \mathcal{Y}_{i,t}}\right) = 0, \tag{21a}$$

$$\mathbb{E}\left(\frac{\partial \mathcal{Y}_{i,t}}{\partial y_t}\right) = 0, \tag{21b}$$

$$Cov\left(\frac{\partial \hat{\sigma}_i^2}{\partial \mathcal{Y}_{i,t}}, \frac{\partial \mathcal{Y}_{i,t}}{\partial y_t}\right) = \frac{2}{\nu_0 - 1}\int_{[0,1]} \mathcal{Y}_{i,t} \cdot \frac{\partial \mathcal{Y}_{i,t}}{\partial y_t}\, di. \tag{21c}$$

The fact that the average marginal change in the estimate of income volatility is zero stems from the symmetry of the Gaussian distribution. As apparent from the estimator in (18), given that symmetry, agents on the opposite sides of the income distribution, but equidistant from its mean, form the same estimate of income volatility. Therefore, the estimates depend, at the margin, on the deviations of income from its mean, $\mathcal{Y}_{i,t}$, that, on average, are null. More formally:

$$\frac{\partial \hat{\sigma}_i^2}{\partial \mathcal{Y}_{i,t}} = \frac{2}{\nu_0 - 1} \cdot \mathcal{Y}_{i,t} \implies \frac{2}{\nu_0 - 1}\int_{[0,1]} \mathcal{Y}_{i,t}\, di = 0.$$

Moreover, the second sufficient statistic (21b) is also zero:

$$\frac{\partial \mathcal{Y}_{i,t}}{\partial y_t} = \frac{\partial y_{i,t}}{\partial y_t} - 1 \implies \int_{[0,1]} \left(\frac{\partial y_{i,t}}{\partial y_t} - 1\right) di = 0.$$

The covariance in (21c) is however different from zero:

$$Cov\left(\frac{\partial \hat{\sigma}_i^2}{\partial \mathcal{Y}_{i,t}}, \frac{\partial \mathcal{Y}_{i,t}}{\partial y_t}\right) = \frac{2}{\nu_0 - 1}\int_{[0,1]} \mathcal{Y}_{i,t} \cdot \left(\frac{\partial y_{i,t}}{\partial y_t} - 1\right) di = \frac{2}{\nu_0 - 1}\int_{[0,1]} (y_{i,t} - y_t) \cdot \frac{\partial y_{i,t}}{\partial y_t}\, di.$$

Its sign depends on the integral term and cannot be established without knowing the values of the individual income sensitivities $(\partial y_{i,t}/\partial y_t)_{i\in[0,1]}$.

3.4.1 Calibration in the simple case

We assume that the true variance of the income process is $\sigma^2 = 0.5$ (following Storesletten et al., 2004) and that average output is equal to 2.[29] Then, we assign a value of $r_t = 4\%$ to the real interest rate and of $\gamma = 3$ to the coefficient of absolute risk aversion.

We can calibrate the hyperparameters by setting, first of all, the prior sample size parameter $\nu_0 > 2$ (recalling that the prior sample size is $n_0 = \nu_0 - 2$), then by choosing the parameter σ_0^2 to target the true value of the fundamental income variance. In particular, we want the prior mean to be equal to the fundamental volatility:[30]

$$\frac{\nu_0 \sigma_0^2}{\nu_0 - 2} = \sigma^2 \implies \sigma_0^2 = \sigma^2 \frac{\nu_0 - 2}{\nu_0}.$$

In our baseline calibration, we set the prior sample size parameter equal to 10, that implies a value for σ_0^2 of 0.40. We simulate a panel of 1,000 agents, assigning to each of them a realization of the income process $y_{i,t} \sim \mathcal{N}(2, 0.5)$ and computing the associated estimates of income volatility according to the formula in (18). See Appendix B.5 for robustness.

In Figure 3a we plot the estimates of the variance of income $\hat{\sigma}_i^2$ as a function of income realizations $y_{i,t}$. We can observe that the estimates are U-shaped, as a consequence of the symmetry of the Gaussian density. Moreover, we can easily observe that the further the income realization is from its mean, the greater the estimate of income volatility, since agents interpret a large deviation from the mean of the income distribution as a signal for larger variance.

Moreover, taking the derivative of the volatility estimator (18), we can find a closed-form expression for the MPCs out of transitory income:

$$MPC_{i,t} = \frac{\partial c_{i,t}}{\partial y_{i,t}} = \frac{1 + r_t}{2 + r_t} - \frac{1}{2 + r_t} \frac{\gamma}{\nu_0 - 1} \cdot (y_{i,t} - y_t). \tag{22}$$

From equation (22) it is clear that MPCs are decreasing in realized income and, when $y_{i,t} < y_t$ ($y_{i,t} > y_t$), they are above (below) the MPC out of wealth.[31] Figure 3b plots the MPCs out of transitory income as a function of earnings, displaying a negative linear relation between MPCs and income. Therefore, our "Inverse-Gamma - Gaussian Bayesian" model is able to reproduce the empirical evidence whereby MPCs decline over the earnings distribution.

This result is an implication of the convexity of the variance estimator (18) in current income (in deviation from its mean). Intuitively, given the normality assumption on the income process (with common and known mean), and given the absence of any persistence in earnings, agents interpret any deviation of their income from its mean as purely transitory and anticipate that mean reversion

[29] We choose this value in such a way to reduce significantly the probability that income realizations take negative values in our simulations.

[30] In this way, there is an anchoring of agents' pre-learning expectations to the true value of the fundamental variance.

[31] Recall that the MPC out of wealth is equal to $(1 + r_t)/(2 + r_t)$.

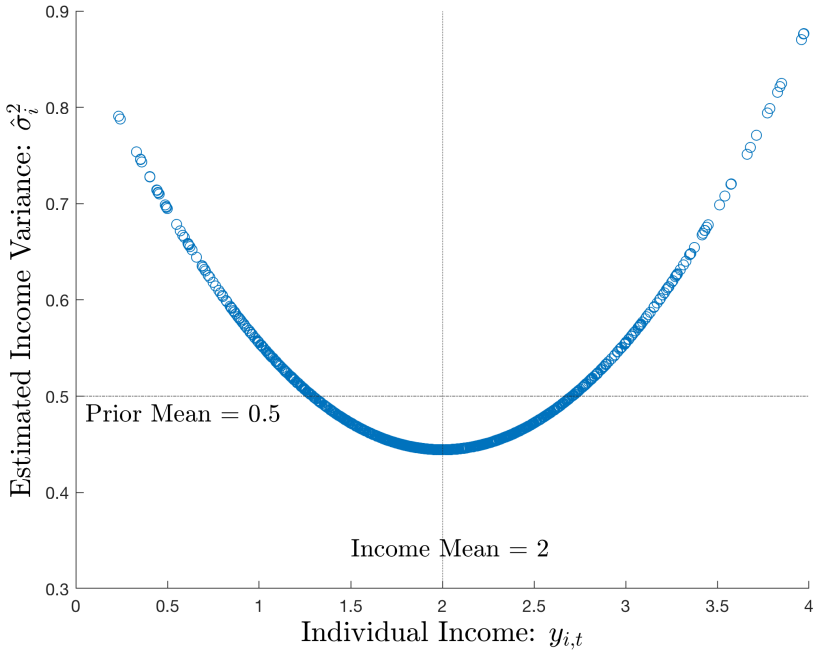

Figure 3a: Heterogeneity in Estimates of Income Volatility over the Income Distribution.

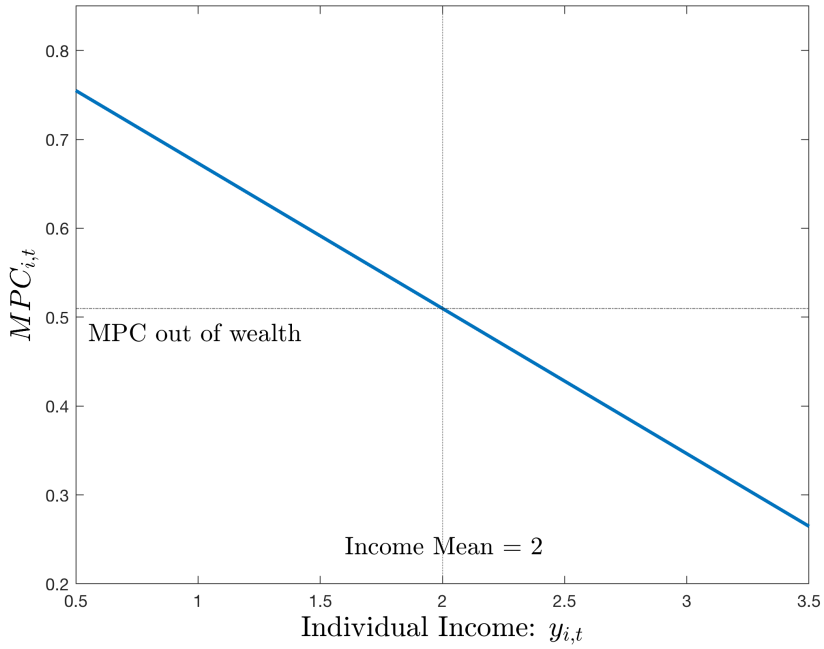

Figure 3b: Heterogeneity in MPCs out of Transitory Income over the Income Distribution.

might take place in the future. Since rich individuals anticipate that their income will revert to the mean in the future, they save a major part of it and have lower MPCs. The opposite is true for poorer agents, who anticipate higher future income and currently spend more.

Suppose for instance, and consistently with the empirical evidence in Patterson (2021), that MPCs correlate positively with income sensitivities, i.e., agents whose earnings are more elastic to aggregate output have higher MPCs. In our model, the positive sign of this correlation is generated by a negative covariance between marginal changes in volatility estimates and earnings sensitivities:

$$Cov\left(\frac{\partial \hat{\sigma}_i^2}{\partial \mathcal{Y}_{i,t}}, \frac{\partial \mathcal{Y}_{i,t}}{\partial y_t}\right) = \frac{2}{\nu_0 - 1} \int_{[0,1]} (y_{i,t} - y_t) \cdot \frac{\partial y_{i,t}}{\partial y_t} \, di \propto -\underbrace{Cov\left(MPC_{i,t}, \frac{\partial y_{i,t}}{\partial y_t}\right)}_{>0} < 0.$$

Under this assumption, poor individuals, who have higher MPCs, have relatively more sensitive earnings to aggregate output. In general equilibrium, this results in amplification of demand shocks on output.

3.5 General equilibrium

We turn next to the characterization of the general equilibrium, by specifying also the *supply side* of the economy and introducing nominal rigidities. We assume that the economy, which runs for two periods $(t, t+1)$, is in the "short-run" at time t and in the "long-run" in time $t+1$.

3.5.1 Supply side

At time t a fraction α of firms (constant and independent of time) keeps its prices unchanged at the predetermined price p^e, which is based on the information set prior to the realization of the shock at time t (Benigno, 2015). A the same time, a fraction $1-\alpha$ can freely adjust the price. In the long-run all prices are perfectly flexible. Therefore α is an implicit index of price rigidity.

The pricing block is described by two equations. First, an equation linking the *optimal relative price* $\tilde{p}_t^* - \tilde{p}_t$ to the the output gap (the deviation of output from the natural flexible-price output):

$$\tilde{p}_t^* - \tilde{p}_t = \zeta \cdot (\tilde{y}_t - \tilde{y}_t^n) \tag{23}$$

where $\zeta > 0$, y_t^n is the natural level of output at time t, and a *tilde* notation denotes variables in deviation from their steady-state values. Hence the optimal price increases relative to the average price whenever economic activity accelerates above the natural level. The second equation of the pricing block expresses the aggregate price level as a weighted-average of the fixed (predetermined) price and the flexible optimized price:

$$\tilde{p}_t = \alpha p^e + (1-\alpha)\tilde{p}_t^* \tag{24}$$

Combining (23) and (24) yields the aggregate supply (AS) relationship:

$$\tilde{p}_t = p^e + \kappa(\alpha) \cdot (\tilde{y}_t - \tilde{y}_t^n) \tag{25}$$

where $\kappa(\alpha) := \alpha^{-1}(1-\alpha)\zeta$ is the slope of the AS curve. Notice that $\kappa(\alpha)' < 0$, i.e., the slope of the supply relation is inversely related to the degree of price rigidity.

3.5.2 Monetary policy

The behavior of monetary policy is summarized by a log-linear interest rate rule whereby the monetary authority reacts to movements in the current price level:

$$\tilde{i}_t = \phi_p \ \cdot \tilde{p}_t + \epsilon_t, \tag{26}$$

where $\phi_p \ > 1$ is a policy feed-back coefficient, and ϵ_t is an iid monetary policy shock.

3.5.3 Steady state

We set government spending in the steady state equal to zero and we normalize aggregate consumption and output to 1. Hence:

$$y = c = 1, \quad g = 0, \quad \tilde{p}_t = p^e = 0.$$

Aggregation implies:

$$\bar{\sigma}^2 := \int_{[0,1]} \hat{\sigma}_i^2 \ di, \tag{27}$$

where $\bar{\sigma}^2$ is defined as the *average* perceived income volatility in the steady state. Moreover, the steady-state value for the real interest rate is determined from the dynamic AD equation (13):

$$\log[\beta(1+r)] = -\frac{\gamma^2}{2}\bar{\sigma}^2. \tag{28}$$

A solution $r > 0$ to this equation exists and is unique. Finally, let us define Γ as a function of the sufficient statistics (i)-(iii) in the steady state:

$$\Gamma := \int_{[0,1]} \left(\frac{\partial \hat{\sigma}_i^2}{\partial y_i} \frac{\partial y_i}{\partial y} \right) di = \mathbb{E}\left(\frac{\partial \hat{\sigma}_i^2}{\partial y_i} \right) \cdot \mathbb{E}\left(\frac{\partial y_i}{\partial y} \right) + Cov\left(\frac{\partial \hat{\sigma}_i^2}{\partial y_i}, \frac{\partial y_i}{\partial y} \right). \tag{29}$$

Notice that $\Gamma = 0$ holds in the two limit cases of a representative-agent and in an homoskedastic version of our model (i.e., $\hat{\sigma}_i = \hat{\sigma}$ for every i).

3.5.4 Log-linear representation

The fully log-linearized model can be described by three equations in the three endogenous variables $(\tilde{y}, \tilde{p}, \tilde{i}_t)$: the AS curve (25), the interest rate rule (26), along with the log-linearized AD equation:[32]

$$\tilde{y}_t = \Theta \cdot \left[\tilde{y}_{t+1} - \frac{1}{\gamma} \cdot \left[\tilde{i}_t - \underbrace{(\tilde{p}_{t+1} - \tilde{p}_t)}_{\text{inflation}} \right] + \tilde{g}_t - \tilde{g}_{t+1} \right], \tag{30}$$

where:

$$\Theta := \left(1 + \frac{\gamma}{2} \Gamma \right)^{-1} > 0, \tag{31}$$

We assume that in time $t+1$ each variable is at its constant natural level, so that $\tilde{y}_{t+1}$=$\tilde{p}_{t+1}$=$\tilde{g}_{t+1}$ =0.

A key feature that distinguishes our framework is the presence, in the AD equation (30), of the parameter Θ, given by the inverse of a linear transformation of Γ, in turn a function of the sufficient statistics (i)-(iii) outlined in Proposition 1. Θ also represents the steady-state output multiplier of government spending of Proposition 1. Notice that parameter $\Theta = 1$ in the standard case of *homoskedasticity* or in the special case of a *representative agent*, and $\Theta > 1$ ($\Theta < 1$) if the average marginal change in risk and/or the covariance sufficient statistics are negative (positive).[33]

3.5.5 The effect of monetary and fiscal shocks

The solution of the system (25), (26), (30) yields the following expression for time t output:

$$\tilde{y}_t = \Omega_t + \frac{\tilde{g}_t - \frac{\epsilon_t}{\gamma}}{\Theta^{-1} + \frac{(1+\phi_p)\kappa(\alpha)}{\gamma}} \tag{32}$$

where

$$\Omega_t := \frac{\frac{1+\phi_p}{\gamma} \kappa(\alpha) \tilde{y}_t^n}{\Theta^{-1} + \frac{(1+\phi_p)\kappa(\alpha)}{\gamma}} \tag{33}$$

The responses of output to an i.i.d. monetary policy shock and to an i.i.d. government spending shock therefore read, respectively:

$$\frac{\partial \tilde{y}_t}{\partial \epsilon_t} = \frac{1}{\frac{\gamma}{\Theta} + (1+\phi_p)\kappa(\alpha)} \tag{34}$$

$$\frac{\partial \tilde{y}_t}{\partial \tilde{g}_t} = \frac{\partial \Omega_t}{\partial \tilde{g}_t} + \frac{1}{\Theta^{-1} + \frac{(1+\phi_p)\kappa(\alpha)}{\gamma}} \tag{35}$$

The role of heteroskedasticity. Notice that, in both cases, the size of the response is

[32] We are also using the Fisher equation (log-linear): $\tilde{r}_t = \tilde{i}_t - (\tilde{p}_{t+1} - \tilde{p}_t)$.

[33] Recall that the income cyclicality effect captured by statistic (ii) is always positive.

increasing in Θ, and therefore decreasing in the hyper-parameter Γ. Relative to either the case of homoskedasticity or the special case of a representative agent economy, the policy shocks lead to amplification or dampening if $\Theta > 1$ or $\Theta < 1$ respectively, which in turn requires $\Gamma < 0$ or $\Gamma > 0$. The amplification case ($\Gamma < 0$), in particular, seems the one empirically most plausible. That case requires two conditions: that the product of the sufficient statistics (i) and (ii) is negative, meaning that on average agents face more risk in the face of negative (policy or demand) shocks to aggregate output; and/or, that the risk-cyclicality covariance statistic (iii) is negative (and large enough in absolute value in the case the product of (i) and (ii) is positive). Recall that the latter statistic describes the interaction between how individual perceived risk varies at the margin with respect to earnings and how sensitive earnings are to aggregate output across the income distribution. It is empirically realistic to assume that the covariance statistic (iii) is negative. That is to say, at the margin, individual perceived risk increases more for those agents whose individual income is more sensitive to aggregate economic conditions. Under those conditions, then, heteroskedasticity (via parameter Θ) acts as an *amplifier* of underlying policy/demand shocks.

The intuition for the above result is simple. Conditional on a negative policy (fiscal or monetary) shock (a fall in $\tilde{g}_t$, or a rise in ϵ_t), if risk increases relatively more for those agents whose individual income is more sensitive to aggregate income, the same individuals will increase their precautionary savings more at the margin, thereby leading to a larger contraction in consumption and output.

4 Conclusion

We have studied the implications of cross-sectional heteroskedasticity in individual income in a tractable heterogeneous agent New Keynesian model. Our setup is grounded in the empirical evidence from micro data showing significant heterogeneity in *perceived* individual income risk over the earnings distribution.

We have proposed *incomplete information* and Bayesian learning as a microfoundation of heteroskedasticity. In our setting, agents cannot observe the true variance of their income process, but estimate it using their income realizations as private signals. Crucially, then, heterogeneity in private signals translates into heterogeneity in the estimated individual income risk. Our model is able to generate heterogeneity in MPCs out of transitory income shocks, and to account for idiosyncratic risk and market incompleteness in an analytically tractable way. Since in our model marginal changes in individual earnings shape income volatility at the margin, saving and consumption choices are also affected at the margin, implying that individual MPCs depend on individual income.

We have shown that, through indirect general equilibrium effects, heteroskedasticity affects the response of output to demand shocks. Those general equilibrium effects hinge on the interaction between the income distribution and changes in individual perceived risk. Most importantly, those effects can be captured by a set of *sufficient statistics* which allow, in principle, to further discipline the model via a small set of moments observable in micro data. For empirically plausible values

of those sufficient statistics, heteroskedasticiy leads to an *amplification* of underlying policy and demand shocks.

We suggest that future research on the macroeconomic implications of cross-sectional heteroskedasticity in individual income should take (at least) three directions. First, empirical estimates of the relevant (three) sufficient statistics should be obtained from micro data. Second, those estimated moments should be used to discipline macroeconomic models with heteroskedasticity in individual income and to understand quantitatively the economic relevance of the heteroskedasticity channel that we propose. Third, the model with incomplete information could be integrated with more elaborated behavioral hypotheses to study how the misperception of the second moment of the income process ("risk") interacts with agents' consumption and saving decisions in general equilibrium.

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APPENDIX

A Empirical estimation

A.1 Dataset

The Panel Survey of Income Dynamics (PSID) is a panel survey, directed by the University of Michigan, that gathers data on demographic and economic characteristics of U.S. families as a whole and on individuals within the family. Data were collected annually from 1968 to 1997 and biennially thereafter. The initial sample in 1968 – of about 5,000 families and 18,000 individuals – consisted of a nationally representative sample and an over-sample of low-income families. The PSID sample has grown over time, as children and grandchildren split-off (i.e., move out from the original family and form a new one). In 1997/1999 and 2017/2019, samples of immigrants have been added. Information is collected through in-person, telephone, and computer-assisted interviewing methods.

We focus on the biannual waves from 1969 to 2019. In particular, we use merged Family files (that contain family-level variables for each wave) from 1969 to 2019 and the Individual file (that contains individual-level variables for each wave, from 1968 to 2019, together with family and individual identifiers over time). We download the data through the "cross-year index" tool available at the link `https://simba.isr.umich.edu/DC/i.aspx` (that allows to directly download selected variables and provide Family and Individual files already merged).

Data cleaning. We restrict the PSID sample to family heads, aged between 25 and 65 years old. First, we remove individuals who misreport age across waves (by more than 2 years from the correct age). We also drop individuals who misreport gender across waves and we drop families that experience a change in the head.[34] Second, we exclude from the analysis individuals with top-coded earnings (0.02% of the sample) and low-income observations (i.e., those with less than $1,000 in 2012 dollars),[35] to focus on individuals with more attachment to the labor force. Then, we keep only individuals who have at least two consecutive waves of (non-missing) earnings (in the trimmed sample).[36] Finally, we drop individuals who report outliers of excessive two-years earnings growth or decline (above 500% or below -300% respectively).[37]

Variables definition. The main variables contained in our final dataset are:

[34] Those families that report at least in one wave *change in family composition* different from "No change in family members," "Change in members other than Head or Wife," or "Head same but Wife left/died and/or Head has new Wife."

[35] Out of 124,947 observations (by observation we mean an individual observed at a given point in time), 14.6% are zero-earnings, and 1.02% have non-zero earnings below $1,000 (2012 dollars). We also exclude very high-income individuals, with more than $1,000,000 in 2012 dollars (0.04% of the sample).

[36] They represent 95% of the trimmed sample.

[37] They are 1.94% of the trimmed sample. Overall, our final sample has 109,688 observations. 89.4% of them are used in the analysis (the remaining ones have either top-coded, zero, below $1,000, or above $1,000,000 earnings).

- *Earnings* (of the household's head): they include labor part of farm income and business income, wages, bonuses, overtime, commissions, professional practice, labor part of income from roomers and boarders or business income. Up to the 1993 wave, earnings include farm income and the labor portion of business income. These are excluded since the 1994 wave. Notice that earnings data in each wave refer to the previous year.
- *Gender* (of the household's head): either male or female.
- *Age* (of the household's head).
- *Change in family composition*: can be "No change in family members," "Change in members other than Head or Wife," "Head same but Wife left/died and/or Head has new Wife."

Each individual is uniquely identified by the *ID* variable. The PCE price index series come from the U.S. Bureau of Economic Analysis.[38] We deflate all nominal variables using the PCE (set to 100 in 2012).

A.2 Estimators and standard errors

Conditional earnings volatilities estimator. To obtain conditional earnings volatilities, first, we residualize (individual) two-years log-earnings – $\log(Y_{it})$ – by regressing them on a constant, a linear time trend, age and gender controls (captured by the vector $Z_{i,t}$):

$$\log(Y_{i,t}) = \mu_g + \delta_g \cdot t + Z'_{i,t} \cdot \zeta + \epsilon_{i,t}, \tag{A-1}$$

where the constant (μ_g) and the coefficients of the time trend (δ_g) are specific to each decile of the lagged earnings distribution, denoted by the subscript g, to which individual i belongs in a given year t. Residuals are denoted by $\epsilon_{i,t}$. The model is estimated with pooled OLS (POLS).

Then, for each year t, we compute the sample variance of residuals within the lagged earnings bin g_{t-1}. Finally, we average these estimates over time. Our conditional variance estimator is, therefore, defined by:

$$Var\left(\epsilon_{i,\cdot}|g\right) = \frac{1}{T}\sum_{t=1}^{T}\Bigg(\underbrace{\frac{1}{|S(g_{t-1})|}\sum_{i\epsilon S(g_{t-1})}\left[\epsilon_{i,t} - \bar{\epsilon}(g_{t-1})\right]^2}_{Var\left(\epsilon_{i,t}|g_{t-1}\right)\text{: conditional variance in year } t}\Bigg), \tag{A-2}$$

where $S(g_{t-1})$ is the set of all individuals who belong to the g-th earnings decile in $t-1$, while $\bar{\epsilon}(g_{t-1})$ is the average of residuals of individuals in $S(g_{t-1})$.[39]

$$S(g_{t-1}) := \{i : \text{earnings group of } i \text{ in } t-1 \text{ is } g\}$$
$$\bar{\epsilon}(g_{t-1}) = \frac{1}{|S(g_{t-1})|}\sum_{i\epsilon S(g_{t-1})}\epsilon_{i,t}$$

[38] They can be downloaded here: PCE.
[39] The symbol $|\cdot|$ denotes the cardinality of the set.

B Model derivations

B.1 Consumer problem

The economy runs for two periods. In the first period t each household can purchase a risk-free (real) bond $a_{i,t+1}$ (paying one unit of good in $t+1$) at the price $1/(1+r_t)$. Additionally, the household can consume a perishable consumption good $c_{i,t}$, using its total income $y_{i,t}$ net of a uniform lump-sum tax T_t,[40] and its initial wealth $a_{i,t}$. In the second period $t+1$ the household consumes its resources entirely. Households maximize their intertemporal utility – discounting with a factor $\beta \in (0,1)$ – choosing consumption and savings subject to two budget constraints.

$$\begin{aligned} \max_{(c_{i,t}, c_{i,t+1}, a_{i,t+1})} \quad & u(c_{i,t}) + \beta \mathbb{E}_{i,t}[u(c_{i,t+1})] \\ s.t \quad & c_{i,t} + \frac{a_{i,t+1}}{1+r_t} = y_{i,t} - T_t + a_{i,t}, \\ & c_{i,t+1} = y_{i,t+1} - T_{t+1} + a_{i,t+1}. \end{aligned}$$

Note that the expectation operator is conditional on agent i's information set. Since agents do not observe the fundamental variance of the income process and they face uncertainty over future income $y_{i,t+1}$, they compute their future expectations with respect to the Gaussian distribution:

$$y_{i,t+1} \mid \mathcal{I}_{i,t} \sim \mathcal{N}(y_{t+1}, \hat{\sigma}_i^2), \tag{B-1}$$

where the relevant variance is the point estimate $\hat{\sigma}_i^2$ obtained in (8) through Bayesian learning.

Consider the usual CARA-exponential utility function: $u(c) = -\frac{1}{\gamma}\exp\{-\gamma c\}$. Then, the Euler equation of the utility maximization problem is:

$$\exp\{-\gamma c_{i,t}\} = \beta(1+r_t) \cdot \mathbb{E}_{i,t}\Big[\exp\{-\gamma c_{i,t+1}\}\Big]. \tag{B-2}$$

Equation (B-2), together with the two budget constraints, characterize the solution of the problem.

To find an explicit solution for consumption, we first take logs of equation (B-2), using the moments of the log-normal distribution, and the two budget constraints to substitute for $c_{i,t+1}$ and

[40] We assume that the government purchases the final good g_t, g_{t+1} and raises taxes with a balanced budget in every period ($\tau = t, t+1$):

$$T_\tau = g_\tau.$$

We could equivalently allow the government to issue bonds in the first period, but this will not change the equilibrium of the model and the results that we obtain.

$a_{i,t+1}$:

$$
\begin{aligned}
-\gamma c_{i,t} &= \log[\beta(1+r_t)] + \log \mathbb{E}_{i,t}\big[\exp\{-\gamma c_{i,t+1}\}\big] \\
-\gamma c_{i,t} &= \log[\beta(1+r_t)] + \log \mathbb{E}_{i,t}\big[\exp\{-\gamma[y_{i,t+1} - T_{t+1} + (1+r_t)(y_{i,t} - T_t + a_{i,t} - c_{i,t})]\}\big] \\
-\gamma c_{i,t} &= \log[\beta(1+r_t)] + \log \mathbb{E}_{i,t}\big[\exp\{-\gamma y_{i,t+1}\}\big] - \gamma[(1+r_t)(y_{i,t} - T_t + a_{i,t} - c_{i,t}) - T_{t+1}] \\
-\gamma c_{i,t} &= \log[\beta(1+r_t)] - \gamma y_{t+1} + \frac{\gamma^2}{2}\hat{\sigma}_i^2 - \gamma[(1+r_t)(y_{i,t} - T_t + a_{i,t} - c_{i,t}) - T_{t+1}]
\end{aligned}
$$

Rearranging the last expression, we find the solution for optimal (current) consumption:

$$
c_{i,t} = -\frac{1}{(2+r_t)\gamma}\log[\beta(1+r_t)] + \frac{1+r_t}{2+r_t}\left[a_{i,t} + y_{i,t} - T_t + \frac{1}{1+r_t}(y_{t+1} - T_{t+1})\right] - \frac{1}{2+r_t}\frac{\gamma}{2}\hat{\sigma}_i^2. \tag{B-3}
$$

B.2 Aggregation and general equilibrium

Let us aggregate the consumption function (B-3) by integrating over the unit interval with respect to the Lebesgue measure to obtain an aggregate demand function:

$$
c_t = -\frac{1}{(2+r_t)\gamma}\log[\beta(1+r_t)] + \frac{1+r_t}{2+r_t}(y_t - g_t) + \frac{1}{2+r_t}(y_{t+1} - g_{t+1}) - \frac{1}{2+r_t}\frac{\gamma}{2}\int_{[0,1]}\hat{\sigma}_i^2 di. \tag{B-4}
$$

Notice that we obtained the previous equation (B-4) by imposing the bond market clearing:

$$
\int_{[0,1]} a_{i,t}\ di = 0,
$$

by using the government budget constraints ($T_\tau = g_\tau$ for $\tau = t, t+1$), and by assuming that individual income integrates to aggregate output:

$$
\int_{[0,1]} y_{i,t}\ di = y_t.
$$

In what follows we postulate that output is endogenous. Moreover, we also assume that when a shock hits the economy, movements in aggregate output shift the entire income distribution with individual earnings sensitivities that are modeled as heterogeneous. More formally, assume that, for every agent $i \in [0,1]$:

$$
y_{i,t} = h_i(y_t, \epsilon_{i,t}),^{41} \tag{B-5}
$$

where $h : \mathbb{R} \to \mathbb{R}$ is some real-valued function, that we assume differentiable and integrable, while $\epsilon_{i,t}$ is an idiosyncratic income shock.

[41] For instance, we can model income realizations as the sum of two components:

$$
y_{i,t} = h_i' \cdot y_t + \epsilon_{i,t},
$$

where $\epsilon_{i,t} \sim \mathcal{N}\big((1-h_i')\cdot y_t, \sigma^2\big)$ and h_i' is the sensitivity of individual to aggregate income. In this way, changes in aggregate output affect individual earnings in a deterministic way. Notice that $y_{i,t} \sim \mathcal{N}\big(y_t, \sigma^2\big)$.

Hence, the sensitivity of individual to aggregate income is:

$$\frac{\partial y_{i,t}}{\partial y_t} = \frac{\partial h_i(y_t, \epsilon_{i,t})}{\partial y_t} =: h_i'. \tag{B-6}$$

Finally, imposing the goods market clearing – $y_t = c_t + g_t$ – in equation (B-4), we obtain a dynamic AD equation:

$$y_t = y_{t+1} - \frac{1}{\gamma}\log[\beta(1+r_t)] - \frac{\gamma}{2}\int_{[0,1]} \hat{\sigma}_i^2 di + g_t - g_{t+1}. \tag{B-7}$$

B.3 Sufficient statistics

We prove Proposition 1.

Proof. Consider a fully transitory government spending shock g_t (i.e., $g_t \neq 0$, $g_\tau = 0$ for $\tau > t$) and let the monetary authority follow the interest rate peg rule (i.e., $r_\tau = r$ for every $\tau \geq t$). Let $\mathcal{H} : \mathbb{R}^2 \to \mathbb{R}$ be the function linking aggregate output and spending (for a given y_{t+1}, g_{t+1}, r_t), defined by:

$$\mathcal{H}(y_t, g_t) = y_t - y_{t+1} + \frac{1}{\gamma}\log[\beta(1+r_t)] + \frac{\gamma}{2}\int_{[0,1]} \hat{\sigma}_i^2 di - g_t + g_{t+1}.$$

Notice that at the solution (y_t, g_t) to the equation $\mathcal{H}(y_t, g_t) = 0$, the dynamic AD equation (13) holds. By the Implicit Function Theorem, if $\mathcal{H} \in \mathcal{C}^1$ (space of continuously differentiable functions) and $\mathcal{H}_y(y_t, g_t) \neq 0$,[42] then there exist two neighborhoods $\mathcal{U}(y_t)$ and $\mathcal{V}(g_t)$ and a function $\tilde{F} : \mathcal{V}(g_t) \to \mathcal{U}(y_t)$ such that:

$$y_t = \tilde{F}(g_t) \iff \mathcal{H}(y_t, g_t) = 0 \quad \forall g_t \in \mathcal{V}(g_t),$$

and:

$$\frac{\partial y_t}{\partial g_t} = \tilde{F}'(g_t) = -\frac{\mathcal{H}_g(y_t, g_t)}{\mathcal{H}_y(y_t, g_t)}.$$

Since:

$$\begin{aligned}
\mathcal{H}_g(y_t, g_t) &= -1, \\
\mathcal{H}_y(y_t, g_t) &= 1 + \frac{\gamma}{2}\frac{d}{dy_t}\int_{[0,1]} \hat{\sigma}_i^2 \; di,
\end{aligned}$$

and assuming that we can exchange the integral and the derivative:[43]

$$\begin{aligned}
\frac{d}{dy_t}\int_{[0,1]} \hat{\sigma}_i^2 \; di &= \int_{[0,1]} \left(\frac{\partial \hat{\sigma}_i^2}{\partial y_{i,t}} \frac{\partial y_{i,t}}{\partial y_t} \right) di = \\
&= \int_{[0,1]} \frac{\partial \hat{\sigma}_i^2}{\partial y_{i,t}} di \int_{[0,1]} \frac{\partial y_{i,t}}{\partial y_t} di + Cov\left(\frac{\partial \hat{\sigma}_i^2}{\partial y_{i,t}}, \frac{\partial y_{i,t}}{\partial y_t} \right),
\end{aligned}$$

[42] With the notation H_y we mean the partial derivative of H to y.

[43] It is sufficient to assume that the function $\hat{\sigma}_i^2 = \sigma^2(y_{i,t})$ is continuously differentiable.

we obtain the output multiplier of government spending:

$$\Theta_t := \frac{\partial y_t}{\partial g_t} = \left\{1 + \frac{\gamma}{2}\left[\mathbb{E}\left(\frac{\partial \hat{\sigma}_i^2}{\partial y_{i,t}}\right) \cdot \mathbb{E}\left(\frac{\partial y_{i,t}}{\partial y_t}\right) + Cov\left(\frac{\partial \hat{\sigma}_i^2}{\partial y_{i,t}}, \frac{\partial y_{i,t}}{\partial y_t}\right)\right]\right\}^{-1}.$$

Following the same steps, we can derive the output multiplier out of a fully transitory interest rate shock:[44]

$$\frac{\partial y_t}{\partial r_t} = -\Theta_t \cdot \frac{1}{\gamma(1 + r_t)}. \tag{B-8}$$

To conclude, since Θ_t depends on the three cross-sectional statistics described in Proposition 1, we can state that these are *sufficient* to characterize the IRFs of aggregate output to demand shocks.

■

B.4 Bayesian updating with Inverse-Gamma

Consider the following model, where the sample $X = (X_1, ..., X_n)$ is i.i.d. from the Gaussian distribution, conditional on the variance σ^2: $X_i \mid \sigma^2 \sim \mathcal{N}(0, \sigma^2)$. Moreover, the prior on σ^2 is an Inverse-Gamma (IG) distribution with hyperparameters $\alpha > 0$, $\beta > 0$, that corresponds to the conjugate prior for the variance of the Gaussian distribution.

$$X_i \mid \sigma^2 \sim \mathcal{N}(0, \sigma^2),$$
$$\sigma^2 \sim \text{Inv-Gamma}(\alpha, \beta).$$

The IG has the following density function $q(\cdot)$:

$$q(\sigma^2) = \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\sigma^2\right)^{-\alpha-1} \exp\left(-\frac{\beta}{\sigma^2}\right),$$

where $\Gamma(\cdot)$ is the Gamma function. The expected value and the variance of the IG distribution are, respectively: $\mathbb{E}(\sigma^2) = \frac{\beta}{\alpha-1}$ and $\text{Var}(\sigma^2) = \frac{\beta^2}{(\alpha-1)^2(\alpha-2)}$.[45]

The likelihood function of the sample $x = (x_1, ..., x_n)$ can be rewritten as:

$$q(x|\sigma^2) \propto \left(\sigma^2\right)^{-n/2} \exp\left(-\frac{n}{2\sigma^2}\frac{1}{n}\sum_{i=1}^{n} x_i^2\right).$$

[44] Equivalently, a shock to v_t.

[45] The existence of these moments requires $\alpha > 1$ and $\alpha > 2$, respectively.

Let us define $w^2 := \frac{1}{n}\sum_{i=1}^{n} x_i^2$ and consider the Bayesian updating stage given the sample x:

$$q(\sigma^2|x) \propto q(x|\sigma^2)\cdot q(\sigma^2) \propto \left(\sigma^2\right)^{-n/2}\exp\left(-\frac{n}{2\sigma^2}w^2\right)\cdot\left(\sigma^2\right)^{-\alpha-1}\exp\left(-\frac{\beta}{\sigma^2}\right) =$$
$$= \left(\sigma^2\right)^{-\alpha-n/2-1}\exp\left(-\frac{\beta + nw^2/2}{\sigma^2}\right).$$

Therefore, the posterior follows an IG distribution:

$$\sigma^2 \mid X = x \sim \text{Inv-Gamma}\left(\alpha + \frac{n}{2}, \beta + \frac{nw^2}{2}\right).$$

Finally, notice that in our tractable example of Section 3.4, $\alpha = \nu_0/2$ and $\beta = \nu_0\sigma_0^2/2$, while $n = 1$ and $w^2 = (y_{i,t} - y_t)^2$.

B.5 Robustness

We study the implications of different values of the hyperparameters of the prior distribution. In particular, we continue to assume that agents set those parameters in such a way to have their prior mean coinciding with the true value of the income variance. In particular, after assigning the prior sample size $n_0 = \nu_0 - 2$, they choose σ_0^2 to have:

$$\sigma_0^2 = \sigma^2\frac{\nu_0 - 2}{\nu_0}.$$

We argue that the prior sample size parameter ν_0 regulates the strength of the belief that agents have in their prior. In fact, from the expression of the variance estimator (18), the greater ν_0, the larger the weight given to the prior mean in the Bayesian updating procedure:

$$\hat{\sigma}_i^2 = \left[\frac{\nu_0 - 2}{\nu_0 - 1}\cdot\underbrace{\frac{\nu_0\sigma_0^2}{\nu_0 - 2}}_{\text{prior mean}} + \frac{1}{\nu_0 - 1}\cdot\underbrace{(y_{i,t} - y_t)^2}_{\text{squared deviation}}\right].$$

Figure B-1 plots the distribution of the estimates of income volatility (as a function of earnings) for different values of the prior sample size hyperparameter.

As the prior sample size increases, the estimates remain U-shaped (since estimates are quadratic in the deviations of the income realization from its mean), but become progressively flatter, converging to the prior mean (and coinciding with the fundamental variance $\sigma_t^2 = 0.5$). Intuitively, the agents give more weight to their prior, while the signal becomes progressively (as ν_0 increases) less relevant for the posterior estimator.[46] Hence, we can approximate the complete information model

[46] On the contrary, if we considered $\nu_0 = 2$, we would be using an improper prior (coinciding with the Jeffreys prior), that is an uninformative one. In this case, not plotted here since the prior mean does not exist, the prior distribution does not give any information to the agent and the variance (posterior) estimator would simply coincide with the

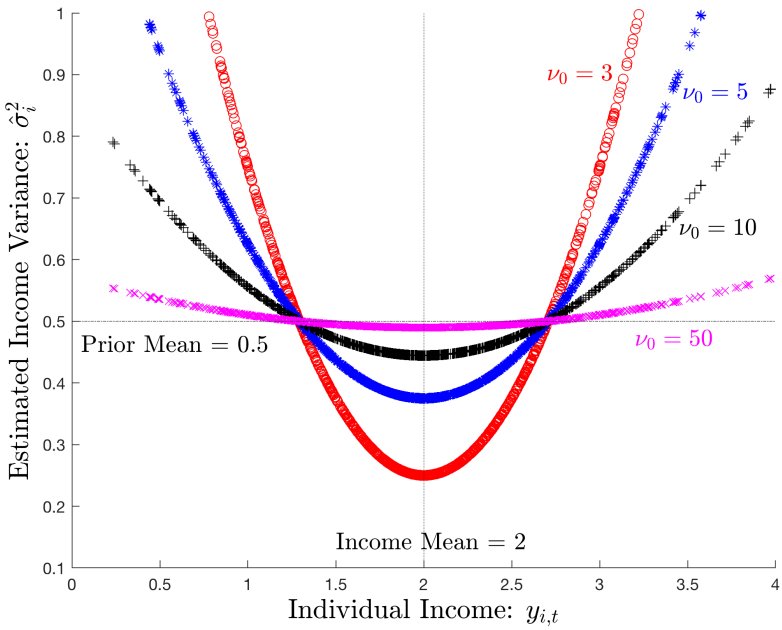

FIGURE B-1: ESTIMATES OF FUTURE INCOME VOLATILITY FOR DIFFERENT PRIOR SAMPLE SIZES.

for large values of the prior sample size, i.e., $\nu_0 \to \infty$:

$$\hat{\sigma}_i^2 = \left[\frac{\nu_0 - 2}{\nu_0 - 1} \cdot \frac{\nu_0 \sigma_0^2}{\nu_0 - 2} + \frac{1}{\nu_0 - 1} \cdot (y_{i,t} - y_t)^2\right] \to \sigma_0^2 = \sigma^2$$

where all agents know that income volatility coincides with the prior mean (i.e., with the true value of the fundamental variance).

squared income deviation from its mean:

$$\hat{\sigma}_i^2 = (y_{i,t} - y_t)^2.$$